

Module 47 - Object Methods & Prototypal Inheritance

1. What are Object Methods?

Object methods are built-in JavaScript functions used to perform different operations on objects, such as creating, copying, accessing, converting, or freezing objects.

2. What is Prototypal Inheritance?

Prototypal Inheritance is a JavaScript mechanism where an object can access properties and methods from another object through its prototype.

Index.js Code:

I. Object.create()

```
let person = { pName: "Johny", age: 30 };
console.log(person);

// Creates a new object whose prototype is 'person'.

let newObj = Object.create(person);
newObj.location = "Tamilnadu";

// Shows the prototype of newObj.

console.log(newObj.proto);

// Gets the prototype of newObj.

console.log(Object.getPrototypeOf(newObj));

// 'age' is inherited from the person object.

console.log(newObj.age);

console.log(newObj);
```

II. Object.assign()

```
let person1 = { id: "QUI123", name1: "Kesavan" };

// Copies the properties into person1 and returns the modified object.

let newObj1 = Object.assign(person1, { role: "Frontend Developer", Salary: 20000 });

console.log(person1); console.log(newObj1);
```

III. Object.entries()

```
let employees = { eName: "Murugan", eRole: "Backend Developer" };  
  
// Converts object properties into an array of key-value pairs.  
  
let multiArr = Object.entries(employees);  
  
multiArr.push(["Name", "Murugesen"]);  
  
console.log(multiArr);
```

IV. Object.fromEntries()

```
// Converts key-value pairs back into an object.  
  
let normalObj = Object.fromEntries(multiArr);  
  
console.log(normalObj);
```

V. Object.keys()

```
// Returns an array containing all property names (keys).  
  
let onlyKeys = Object.keys(normalObj);  
  
console.log(onlyKeys);
```

VI. Object.values()

```
// Returns an array containing all property values.  
  
let onlyValues = Object.values(normalObj);  
  
console.log(onlyValues);
```

VII. Object.freeze()

```
let newObj2 = { id: 1 };  
  
// Prevents adding, deleting, or modifying properties.  
  
Object.freeze(newObj2);  
  
newObj2.name1 = "Javascript";  
  
newObj2.id = "QUI123";  
  
console.log(newObj2);
```

VIII. Object.isFrozen()

```
// Checks whether the object is frozen.  
  
console.log(Object.isFrozen(newObj2));  
  
console.log(Object.isFrozen(newObj1));  
  
console.log(Object.isFrozen(normalObj));
```

Index.html Code:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Object Methods, Prototypal Inheritance</title>  
</head>  
<body>  
  <h1>Object Methods - Prototypal Inheritance</h1>  
  <script src="index.js"></script>  
</body>  
</html>
```